20/09/2026

Code-E

Aakash
Medical | IIT-JEE | Foundations

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Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **AIATS Prime for NEET-2027(XII Passed)_Unit Test-05** **Time : 180 Min.**

Topics Covered:
Physics : Thermal Properties of Matter, Thermodynamics, Kinetic Theory
Chemistry : Solutions, Chemical Kinetics
Botany : Respiration in Plants, Plant Growth and Development
Zoology : Animal Kingdom (General Account & Non chordates), Animal Kingdom (Chordates)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. A body cools from a temperature $3T$ to $2T$ in 10minutes. The room temperature is T. Assume that Newton's law of cooling is applicable. The temperature of the body at the end of next 10 minutes will be

(1) $\frac{7}{4}T$

(2) $\frac{3}{2}T$

(3) $\frac{4}{3}T$

(4) T

2. Time period of a pendulum is 4 s and coefficient of linear expansion of material of its string is $2 \times 10^{-3}\ K^{-1}$. If temperature is increased by 50°C, then change in time per second is

(1) 0.2 s

(2) 0.02 s

(3) 0.05 s

(4) 0.1 s

3. The unit of thermal conductivity is

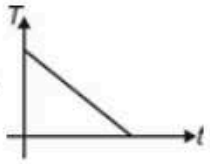
(1) $J\,m^{-1}K^{-1}$

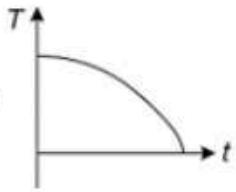
(2) $J\,m^{-2}K^{-1}$

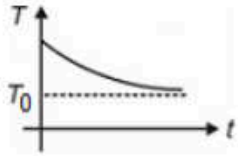
(3) $W\,m^{-1}K^{-1}$

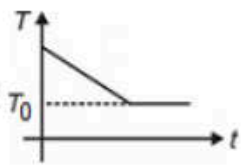
(4) $W\,m\,K^{-1}$

4. An object is allowed to cool in a colder surroundings at temperature T_0. If T is the temperature of object, the correct variation of temperature T with time t is

(1) 

(2) 

(3) 

(4) 

5. For perfectly black body emissivity (e) is

(1) $0 < e < 1$

(2) $e = 0$

(3) $e > 1$

(4) $e = 1$

6. Two metal spheres of same material at same temperature have radius in the ratio 1 : 3. The ratio of their rate of cooling is

(1) 9 : 1

(2) 3 : 1

(3) 81 : 1

(4) 27 : 1

7. Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which of the following option is correct.

(1) $\alpha_1 \ell_2 = \alpha_2 \ell_1$

(2) $\alpha_1 \ell_1^2 = \alpha_2 \times \ell_2^2$

(3) $\alpha_1 \ell_1 = \alpha_2 \ell_2$

(4) $\alpha_1^2 \ell_2 = \alpha_2^2 \times \ell_1$

8. 10 gram of ice at 0 °C is added to 30 g of water at 20 °C. If the latent heat of ice is 80 cal/g, then the final temperature of mixture is

(1) –5 °C

(2) 0 °C

(3) +5 °C

(4) 2 °C

9. The rectangular surface of area $a \times b$ of a black body at a temperature of $400\ \mathrm{K}$ emits energy at the rate E per second. If the length and breadth of the surface are reduced to half of the initial value and the temperature is raised to $600\ \mathrm{K}$, then the new rate of emission of energy will be

(1) $\frac{81}{64}E$

(2) $\frac{81}{32}E$

(3) $\frac{27}{64}E$

(4) $\frac{27}{32}E$

10. A bucket full of hot water is kept in a room and it cools from 75°C to 70°C in T_1 minute and from 70°C to 65°C in T_2 minute and from 65°C to 60°C in T_3 minute then

(1) $T_1 = T_2 = T_3$

(2) $T_1 < T_2 < T_3$

(3) $T_1 > T_2 > T_3$

(4) $T_1 < T_2 > T_3$

11. The power radiated by a blackbody is P at a certain temperature. If power radiated by this body becomes $\frac{1}{16}\mathrm{th}$ times, then wavelength corresponding to maximum intensity becomes

(1) 2 times

(2) $\frac{1}{2}$ times

(3) 4 times

(4) $\frac{1}{4}$ times

12. Three slabs of thermal resistance R, $2R$ and $3R$ are connected in parallel. The equivalent thermal resistance of the combination is

(1) $6R$

(2) $\frac{6R}{11}$

(3) $\frac{11R}{6}$

(4) $\frac{3R}{5}$

13. If temperature of an object at Fahrenheit scale is 104°F, then its temperature in centigrade scale is

(1) 40°C

(2) 136°C

(3) 76°C

(4) 104°C

14. A system is taken from state A to state B along two different paths 1 and 2. If the heat absorbed and work done by the system along these two paths are Q_1, Q_2 and W_1, W_2 respectively, then

(1) $Q_1 = Q_2$

(2) $W_1 = W_2$

(3) $Q_1 - W_1 = Q_2 - W_2$

(4) $Q_1 + W_1 = Q_2 + W_2$

15. In the range 273 K to 277 K, the correct statement(s) is/are

(1) Water expands on heating

(2) Water contracts on cooling

(3) Maximum density of water occurs at 277 K

(4) All of these

16. **Statement A :** Two bodies are said to be in thermal equilibrium when no (net) heat flows from one body to other.
Statement B: At thermal equilibrium, two bodies can be at different temperature.

(1) Statement A is true but statement B is false

(2) Statement A is false but statement B is true

(3) Both statements A and B are true

(4) Both statements A and B are false

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17. For a gaseous state if heat supplied to the system is 100 J and work done by the system is 25 J, then internal energy of the system is

(1) 125 J

(2) 100 J

(3) 50 J

(4) 75 J

18. Which of the following ratio is correct for isobaric process?

(1) $\Delta Q : \Delta U = 1 : 1$

(2) $\Delta Q : W = (\gamma - 1) : \gamma$

(3) $\Delta Q : \Delta U = \gamma : 1$

(4) $\Delta Q : W = 1 : \gamma$

19. **In the following question, a statement of assertion (A) is followed by a statement of reason (R)**
A : Internal energy of a gas must increase when its temperature is raised.
R : Whenever heat is supplied to the gas its internal energy always increases.

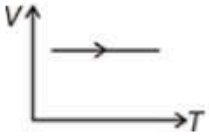
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

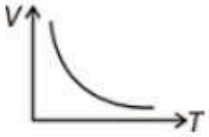
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

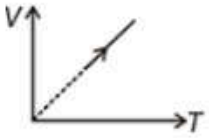
(3) Assertion is true statement but Reason is false

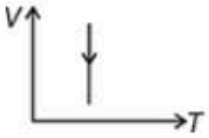
(4) Both Assertion and Reason are false statements

20. *V-T* graph for isobaric process is
(Where symbols have their usual meaning)

(1) 

(2) 

(3) 

(4) 

21. For an adiabatic process which among the following is correct? $\left(\gamma = \frac{C_p}{C_V}\right)$

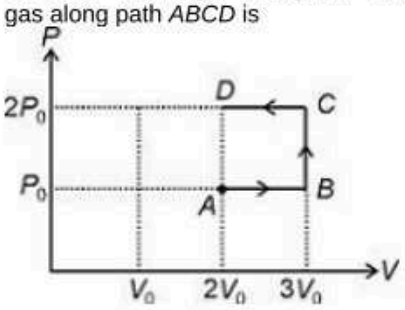
(1) $T^{\gamma-1}V = \text{constant}$

(2) $T^{\frac{1}{\gamma-1}}V = \text{constant}$

(3) $TV^{\gamma} = \text{constant}$

(4) $T^{\gamma-1}V^{\gamma} = \text{constant}$

22. Consider following *PV*-diagram. Work done by an ideal gas along path *ABCD* is



(1) P_0V_0

(2) $-P_0V_0$

(3) $2P_0V_0$

(4) $4P_0V_0$

23. An ideal gas obeys the law P^3V = constant. The initial temperature and pressure are T_0 and P_0 respectively. If the gas is compressed to pressure $3P_0$ then final temperature will be

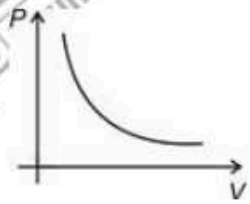
(1) $9T_0$

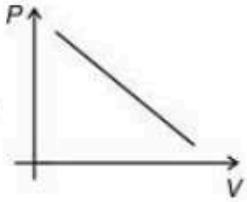
(2) $\frac{T_0}{9}$

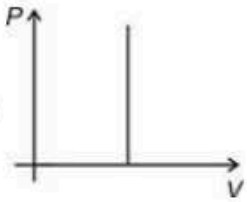
(3) $\frac{T_0}{3}$

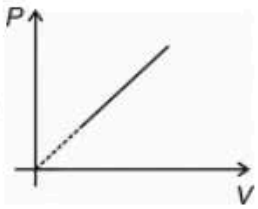
(4) $\frac{T_0}{\sqrt{3}}$

24. Which of the following *P-V* diagram best represents the isothermal process?

(1) 

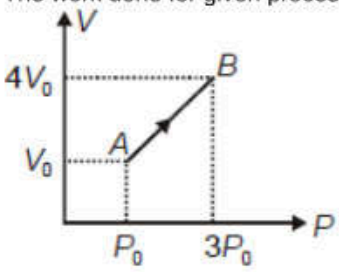
(2) 

(3) 

(4) 

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25. The work done for given process *AB* is



(1) $6\ P_0V_0$

(2) $5\ P_0V_0$

(3) $3\ P_0V_0$

(4) $2P_0V_0$

26. One mole of an ideal gas expands from volume *V* to 2*V* at a constant temperature *T* K. Heat absorbed by the gas is

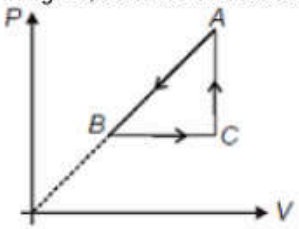
(1) *RT* ln 2

(2) *RT* ln 0.5

(3) *RT*

(4) $\frac{RT}{2}$

27. If the *P-V* diagram of a cyclic process *ABCA* is as shown in figure, then the correct statement is



(1) $\Delta Q_{A\to B} = \text{negative}$

(2) $\Delta U_{B\to C} = \text{positive}$

(3) $W_{CAB} = \text{negative}$

(4) All of these

28. A sample of gas expands from volume V_1 to V_2. The amount of work done by the gas is minimum when the expansion is

(1) Adiabatic

(2) Isobaric

(3) Isothermal

(4) Equal in all cases

29. 70 kcal heat is absorbed by an ideal diatomic gas heated at constant pressure. Change in internal energy of the gas is

(1) 50 kcal

(2) 70 kcal

(3) 20 kcal

(4) 42.5 kcal

30. The surface of a lake is at –1°C. Then temperature of bottom of deep lake is

(1) –1°C

(2) 0°C

(3) 4°C

(4) 10°C

31. The maximum efficiency of an engine working between temperatures 100 K and 400 K is

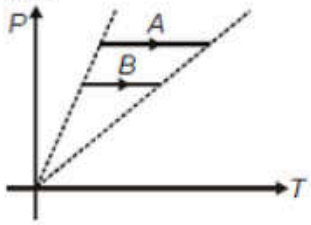
(1) 75%

(2) 50%

(3) 100%

(4) 25%

32. Pressure temperature (*P-T*) graph for two processes *A* and *B* on a system is shown in figure. If W_1 and W_2 are work done by the gas in process *A* and *B* respectively, then



(1) $W_1 = W_2$

(2) $W_1 < W_2$

(3) $W_1 > W_2$

(4) $W_1 = -W_2$

33. Efficiency of heat engine working between 300 K and 600 K cannot be

(1) 0.3

(2) 0.4

(3) 0.5

(4) 0.6

34. Mean translational kinetic energy per molecule for diatomic ideal gas is (k_B : Boltzmann constant)

(1) $\frac{3}{2}RT$

(2) $\frac{3}{2}k_BT$

(3) $\frac{5}{2}RT$

(4) $\frac{1}{2}k_BT$

35. The density of a gas is 6×10^{-2} kg/m^3 and root mean square velocity of the gas molecules is 500 m/s. The pressure exerted by the gas on the walls of the vessel is

(1) 5×10^3 Pa

(2) 1.2×10^{-4} Pa

(3) 0.83×10^{-4} Pa

(4) 30 Pa

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36. A gas has molar heat capacity $C = \frac{7R}{2}$ in the process PT = constant. The number of degrees of freedom of molecules in the gas is

(1) 1

(2) 2

(3) 3

(4) 4

37. The degree of freedom of mixture of two ideal gases having moles n_1 and n_2 and degree of freedom f_1 and f_2 respectively, is

(1) $f_{mix} = \frac{n_1 f_1 + n_2 f_2}{\frac{1}{n_1} + \frac{1}{n_2}}$

(2) $f_{mix} = \frac{n_1 f_2 + n_2 f_1}{n_1 + n_2}$

(3) $f_{mix} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$

(4) $f_{mix} = \frac{\frac{n_1}{f_1} + \frac{n_2}{f_2}}{n_1 + n_2}$

38. 3 moles of monatomic gas at temperature T is mixed with 5 moles of diatomic gas at temperature $3T$. If there is no loss of heat to surroundings, the temperature of mixture after some time is

(1) $2T$

(2) $\frac{17T}{42}$

(3) $\frac{33T}{8}$

(4) $\frac{42T}{17}$

39. The ratio of rms speed and most probable speed for a diatomic gas molecule (respectively) is

(1) $\sqrt{\frac{5\pi}{8}}$

(2) $\sqrt{\frac{3\pi}{8}}$

(3) $\sqrt{\frac{5}{2}}$

(4) $\sqrt{\frac{3}{2}}$

40. Consider the statements regarding the mean free path of gas molecules.
(a) It is inversely proportional to number density of gas molecules.
(b) It is inversely proportional to pressure of gas.
(c) It is inversely proportional to square of diameter of gas molecule.
Based on above information pick the correct option.

(1) Only (a) and (b) are correct

(2) Only (a) is correct

(3) Only (a) and (c) are correct

(4) All (a), (b) and (c) are correct

41. The average value of which of the following quantities is zero for the molecules of an ideal gas in equilibrium?

(1) Kinetic energy

(2) Velocity

(3) Density

(4) Speed

42. Ratio of molar specific heats at constant pressure and that at constant volume for N_2 gas is

(1) $\frac{5}{2}$

(2) $\frac{7}{3}$

(3) 2

(4) $\frac{7}{5}$

43. Match Column I and Column II and choose the correct option from the given choices. (Symbol have their usual meaning)

	Column I		Column II
a.	Root mean square speed of gas molecules	(i)	$\frac{1}{3}\rho V_{rms}^2$
b.	Pressure exerted by ideal gas	(ii)	$\sqrt{\frac{3RT}{M}}$
c.	Average translational kinetic energy of a gas molecule	(iii)	$\frac{5}{2}RT$
d.	Total internal energy of 1 mole of a diatomic gas	(iv)	$\frac{3}{2}k_BT$

(1) a(iii), b(ii), c(i), d(iv)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(ii), b(i), c(iv), d(iii)

44. A gas at 250 K is confined in a closed container. If the temperature of the gas is raised by 5 K, then the percentage increase in pressure will be

(1) 2%

(2) 3%

(3) 4%

(4) 5%

45. **Statement A :** A process in which the temperature of the system is kept fixed throughout is called an isobaric process.
Statement B : If the system is insulated from the surroundings and no heat flows between the system and the surroundings, the process is adiabatic.
Statement C : Internal energy of a system is the sum of kinetic energies and potential energies of the molecular constituents of the system.
In light of above statements, choose the correct option:

(1) Only statement A is true

(2) Only statement B is true

(3) Both statements B and C are true

(4) Only statement C is true

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CHEMISTRY

46. If the mass ratio of methanol and water in a solution is 4 : 9 then the mole fraction of methanol in the solution will be

(1) 0.4

(2) 0.1

(3) 0.6

(4) 0.2

47. K_H values for some gases at the same temperature T are given below

Gas	K_H/kbar
A	30
B	25
C	47
D	20

Where K_H is Henry's law constant in water. The order of their solubility in water is

(1) A > B > C > D

(2) D > C > B > A

(3) D > B > A > C

(4) C > A > B > D

48. For an ideal solution which thermodynamic relation is incorrect?

(1) $\Delta_{mix}H = 0$

(2) $\Delta_{mix}V = 0$

(3) $\Delta_{mix}S = 0$

(4) $\Delta_{mix}G < 0$

49. What will be the concentration of chloride ion if 200 mL of 0.2 M $CaCl_2$ is mixed with 300 mL of 0.1 M NaCl solution?

(1) $\frac{11}{5}$M

(2) $\frac{11}{50}$M

(3) $\frac{22}{49}$M

(4) $\frac{1}{50}$M

50. Consider the following statements
(a) Solubility of gases in liquids decreases with rise in temperature.
(b) At high altitudes the partial pressure of oxygen is less than that at the ground level
(c) Henry's law constant (K_H) is a function of the nature of the gas.
The **correct** statement are :

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

51. If the density of H_2SO_4 solution is 1.84 gm/cc and contains solute 49% by mass, then the molarity of H_2SO_4 solution is

(1) 18.4

(2) 10.8

(3) 9.2

(4) 4.9

52. Consider the given solutions of
(a) Ethanol and acetone
(b) Phenol and aniline
(c) Carbon disulphide and acetone
(d) Chloroform and acetone
(e) Bromoethane and chloroethane
The solutions which exhibit positive deviation from Raoult's law are

(1) (a), (b) and (c) only

(2) (a) and (c) only

(3) (b), (d) and (e) only

(4) (c) and (e) only

53. The mixture that forms minimum boiling azeotrope is

(1) Hydrochloric acid + Water

(2) Water + Nitric Acid

(3) Water + Ethanol

(4) Acetone + Aniline

54. The mass of urea required in making 4 kg of 0.4 molal aqueous solution is

(1) 96 g

(2) 93.75 g

(3) 90 g

(4) 100 g

55. A solution is made by two volatile components A $\left(P^{\circ}_{A} = 100 \text{ mm Hg}\right)$ and B $(P^{\circ}_{B} = 200 \text{ mm Hg})$. Find out the mole fraction of A in vapour phase if solution is made by mixing equal moles of A and B.

(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{2}{3}$

(4) $\frac{1}{4}$

56. Upon mixing of acetone and ethanol, which of the following is not observed?

(1) $\Delta H_{mix} > 0$

(2) $\Delta S_{mix} > 0$

(3) $\Delta G_{mix} > 0$

(4) $\Delta V_{mix} > 0$

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57. Consider the following statements labelled as assertion (A) and Reason (R).
Assertion (A) : Aquatic species are more comfortable in cold water rather than in warm water.
Reason (R) : Solubility of a gas in a liquid increases with decrease in temperature.
In the light of above statements, choose the correct option.

(1) Assertion and reason both are correct statements and reason is correct explanation for assertion.

(2) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

(3) Assertion is correct statement but reason is wrong statement.

(4) Assertion is wrong statement but reason is correct statement.

58. Select the process in which final vapour pressure is more than initial vapour pressure?

(1) 0.01 mol NaCl added to 0.01 M NaCl(aq)

(2) 0.01 mol sugar added to 100 ml H_2O

(3) 100 mole H_2O added to 100 ml 0.01 M KCl (aq) solution

(4) 0.1 mol sugar added to 0.1 M NaCl solution

59. Boiling point of 0.1m urea(aq) solution is ($K_b(H_2O)$ = 0.52 K molal^{-1})

(1) 100°C

(2) 101°C

(3) 100.052°C

(4) 101.052°C

60. The value of van't Hoff factor is maximum for (consider complete dissociation of solute)

(1) NaCl

(2) $MgSO_4$

(3) K_2SO_4

(4) KCl

61. 0.1 molal aqueous solution of which compound will have highest boiling point? (consider complete dissociation of electrolytes)

(1) Glucose

(2) KCl

(3) $CaCl_2$

(4) CH_3COONa

62. The van't Hoff factor for 0.1 M $Ba(NO_3)_2$ is 2.74. Percentage degree of dissociation is

(1) 91.3%

(2) 87%

(3) 75%

(4) 63.5%

63. Match solutions in List-I with their osmotic pressure at 27°C in List-II.

	List-I		List-II
a.	0.1 M Na_2CO_3 with 40% ionisation	(i)	3.69 atm
b.	0.05 M NaCl with 80% ionisation	(ii)	7.38 atm
c.	0.3 M urea	(iii)	2.21 atm
d.	0.15 M glucose	(iv)	4.43 atm

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(i), b(iii), c(ii), d(iv)

(4) a(iv), b(ii), c(iii), d(i)

64. 0.004 M solution of Na_2SO_4 is isotonic with a 0.0108 M solution of glucose at same temperature. The degree of dissociation of Na_2SO_4 is

(1) 75%

(2) 50%

(3) 25%

(4) 85%

65. Match List-I with List-II.

	List-I [Electrolyte with degree of dissociation (α)]		List-II [van't Hoff factor (i)]
a.	Na_2CO_3 ; α = 70%	(i)	3.4
b.	$K_4[Fe(CN)_6]$; α = 60%	(ii)	1.8
c.	$Al_2(SO_4)_3$; α = 75%	(iii)	2.4
d.	KCl; α = 80%	(iv)	4.0

The correct option is

(1) a(i), b(ii), c(iv), d(iii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(iii), b(i) , c(iv), d(ii)

66. For the reaction, 3A + 2B + C→ Products, the following kinetic data was obtained

Initial concentration			Initial rate
A(mol L^{-1})	B(mol L^{-1})	C(mol L^{-1})	(mol L^{-1} s^{-1})
0.01	0.005	0.01	5×10^{-5}
0.02	0.005	0.01	5×10^{-5}
0.01	0.010	0.01	2.5×10^{-5}
0.01	0.005	0.02	20×10^{-5}

Hence, the rate of reaction is given by

(1) Rate = k [A] [B] [C]

(2) Rate = k $[A]^0$ $[B]^{-1}$ $[C]^2$

(3) Rate = k [A] [B] $[C]^{-2}$

(4) Rate = k [A] [B] $[C]^2$

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67. A hypothetical reaction, $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below

$A_2 \rightleftharpoons A + A$ (fast)

$A + B_2 \rightarrow AB + B$ (slow)

$A + B \rightarrow AB$ (fast)

The overall order of the reaction is

(1) 2

(2) 1

(3) 1½

(4) 0

68. For the reaction, $3A + 2B \rightarrow 5C + 4D$. Which of the following does not express the reaction rate?

(1) $\frac{1}{3}\frac{d[A]}{dt}$

(2) $\frac{1}{5}\frac{d[C]}{dt}$

(3) $\frac{1}{4}\frac{d[D]}{dt}$

(4) $-\frac{1}{2}\frac{d[B]}{dt}$

69. Match order of reaction in list I with their unit of rate constant in list II

	List I		List II
a.	Zero order reaction	(i)	$Mol^{-1} L s^{-1}$
b.	First order	(ii)	$Mol L^{-1} s^{-1}$
c.	Second order	(iii)	$Mol^{-2} L^2 s^{-1}$
d.	Third order	(iv)	s^{-1}

Correct match is

(1) a(iii), b(i), c(ii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(ii), b(iv), c(iii), d(i)

70. Half life of a zero order reaction is 15 min. Time required to consume reactant completely will be

(1) 20 minutes

(2) 45 minutes

(3) 30 minutes

(4) 60 minutes

71. The energies of activation for forward and reverse reactions for $X_2 + Y_2 \rightleftharpoons 2XY$ are 140 kJ mol^{-1} and 180 kJ mol^{-1} respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reaction by 100 kJ mol^{-1}. The enthalpy change of the reaction (X_2 + $Y_2 \rightarrow 2XY$) in the presence of catalyst will be in (kJ mol^{-1})

(1) -40 kJmol^{-1}

(2) -140 kJmol^{-1}

(3) $+40$ kJmol^{-1}

(4) $+140$ kJmol^{-1}

72. For an elementary reaction, $A + 2B \rightarrow 2C + D$ the molecularity is

(1) Zero

(2) One

(3) Two

(4) Three

73. The $t_{1/2}$ of first order reaction is 40 minutes. The percentage of reactant left after 120 minutes is

(1) 75%

(2) 50%

(3) 12.5%

(4) 6.25%

74. For a certain reaction the rate = $k[A]^2[B]^{-1}$ if concentration of $[A]$ is doubled and $[B]$ is halved, then rate becomes

(1) 2 times to the initial rate

(2) 4 times to the initial rate

(3) 6 times to the initial rate

(4) 8 times to the initial rate

75. Which among the following is an **incorrect** statement?

(1) Order of a reaction is an experimental quantity

(2) Order can be zero but cannot be fractional

(3) Order is applicable to elementary as well as complex reactions

(4) For complex reaction, order is given by the slowest step of the reaction

76. The rate constant of the given reaction,

$R(g) \xrightarrow{\Delta} 2P(g)$ is 2.48×10^{-4} s^{-1}

A 1 : 1 molar ratio of R to P in the reaction mixture is attained after

(1) 32 min

(2) 27.3 min

(3) 20 min

(4) 0 min

77. A reaction proceeds by first order, 75% of this reaction was completed in 32 min. The time required for 50% completion is

(1) 8 min

(2) 16 min

(3) 20 min

(4) 24 min

78. Slope of the plot ln k versus $\frac{1}{T}$ for Arrhenius equation is

(1) ln A

(2) $-\frac{Ea}{R}$

(3) $\frac{Ea}{R}$

(4) $\frac{Ea}{A}$

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79. If the rate of reaction increases to two times with every 10° rise in temperature, and the temperature is raised by 60°, the new rate of the reaction will be

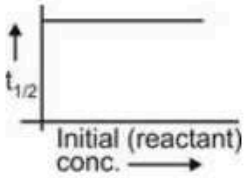
(1) 4 times

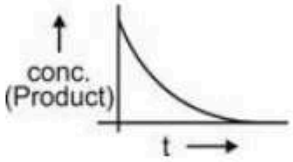
(2) 64 times

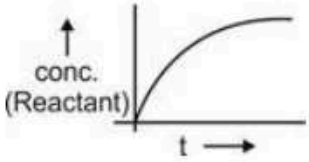
(3) 32 times

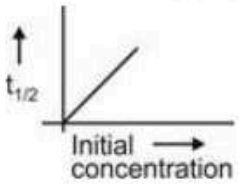
(4) 16 times

80. Select the correct graph for 1st order reaction.

(1) 

(2) 

(3) 

(4) 

81. The rate constant for a chemical reaction taking place at $500\ \mathrm{K}$ is expressed as $\mathrm{k} = \mathrm{Ae}^{-1000}$. The activation energy of reaction is

(1) 5000 R

(2) 5 x 10^4 R

(3) 5 x 10^5 R

(4) 2000 R

82. 75% of a first-order reaction is completed in 30 minutes. The time required for completion of 93.75% of the reaction (in minutes) is

(1) 120

(2) 45

(3) 60

(4) 90

83. The rate of a reaction becomes quadruples when temperature changes from 303 K to 323 K. The energy of activation of the reaction will be (Assuming energy of activation does not change with temperature)

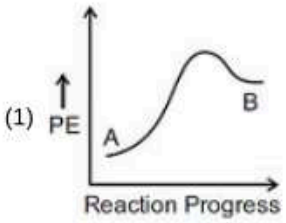
(1) 56.4 kJ/mol

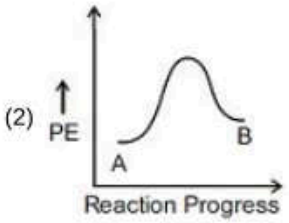
(2) 28.2 kJ/mol

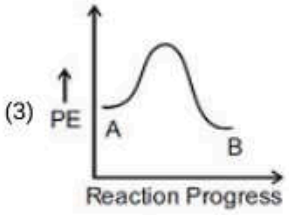
(3) 102.8 kJ/mol

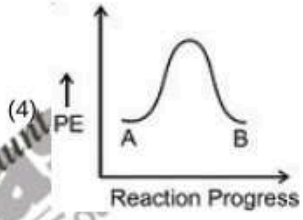
(4) 92.8 kJ/mol

84. For a reaction A → B, enthalpy of reaction is –4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1}. The correct potential energy profile for the reaction is shown in option.

(1) 

(2) 

(3) 

(4) 

85. Given below are two statements

Statement I : A catalyst provides an alternate pathway or reaction mechanism by reducing the activation energy between reactants and products.

Statement II : Catalyst catalyses the spontaneous reactions but does not catalyse non-spontaneous reactions.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

86. The rate of first-order reaction is 0.05 mol L^{-1} S^{-1} at 10 seconds and 0.04 mol L^{-1} S^{-1} at 20 seconds after the initiation of the reaction. The half-life period of the reaction is nearly (log5 = 0.69)

(1) 24.36 s

(2) 40.45 s

(3) 31.07 s

(4) 90.23 s

87. If rate constant and rate of reaction both have same units, then the order of reaction is

(1) 0

(2) 1

(3) 2

(4) 3

88. For a radioactive reaction, 90% of the reaction completes in 10 years. What will be its half-life?

(1) 10 years

(2) 5 years

(3) 7 years

(4) 3 years

89. Observed molar mass of acetic acid which is dimerizes 75% in benzene solvent, is

(1) 60

(2) 96

(3) 37.5

(4) 30

90. Molality of an aqueous solution in which mole fraction of glucose is 0.1, is

(1) 6.17 m

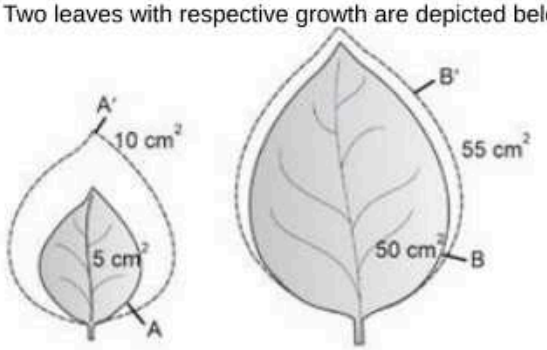
(2) 0.72 m

(3) 2.17 m

(4) 12.34 m

BOTANY

91. Two leaves with respective growth are depicted below



Considering A′ and B′ as final size after growth, which of the following statements explain the above situation appropriately?

(1) Relative growth rate of leaf A′ is twice as that of B′ relative growth rate.

(2) Absolute growth rate is same in A′ and B′.

(3) Absolute growth rate of B′ is 1/10 of absolute growth rate of A′.

(4) Relative growth rate of B′ is same as relative growth rate of 'A'.

92. Read the following statements carefully.
a. The brewing industry benefits from this hormone, as it increases the rate of production of fermentable sugars from seeds.
b. The spraying of this PGR on sugarcane plants promotes internode elongation.
c. This PGR is applied to induce thinning in cotton.
d. This PGR can be isolated from natural sources such as a parasitic fungus on rice plant.
How many of the above statements is/are **true** regarding Gibberellic acids?

(1) One

(2) Two

(3) Three

(4) Four

93. Read the following statements and state **True (T)** or **False (F)**.
A. Relative growth rate is the measurement and comparison of total growth per unit time.
B. Primary growth is a characteristic of all types of plants.
C. Auxin, ethylene and ABA are the most common plant growth inhibitors.
D. Cytokinins delay cell growth while auxins are involved only in cell division.

(1) A(T), B(T), C(F), D(F)

(2) A(F), B(T), C(F), D(F)

(3) A(F), B(F), C(T), D(F)

(4) A(F), B(T), C(F), D(T)

94. Which among the following is **not** the characteristic of cells in meristematic phase of growth?

(1) Rich in protoplasm

(2) Increased vacuolation

(3) Possess large conspicuous nuclei

(4) Abundant plasmodesmatal connections

95. Find the **correct** application of cytokinin.

(1) Acts as antitranspirant and induces closure of stomata

(2) To promote root development on various types of cuttings

(3) In tea plantation for giving bushy appearance

(4) Promotes parthenocarpy in fruits like banana and grapes

96. Which of the following plant hormones is **not** acidic in nature?

(1) Gibberellin

(2) Auxin

(3) Cytokinin

(4) ABA

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97. Consider the following Assertion and Reason and choose the **correct** option.
Assertion (A) : Development in plants is considered as the sum of growth and differentiation
Reason (R) : Development in plants is under the control of intrinsic factors only.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

98. If we apply ethylene on cucumber plants, the fruit yield increases. This is due to

(1) Decrease in the rate of catabolic processes in the plant

(2) Reduction in the requirement of minerals by stems and leaves

(3) Increase in the water retaining capacity of the plant

(4) Increase in the number of female flowers in the plant

99. Consider the following statements.
Statement A: During plant growth, dedifferentiation is pre-requisite for re-differentiation.
Statement B: Growth in higher plants is open while differentiation is closed.
In the light of above statements, choose the **correct** option

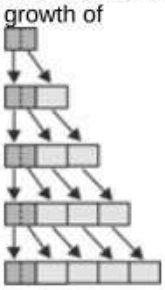
(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statement A and statement B are correct

(4) Both statement A and statement B are incorrect

100. The below given growth pattern can be used to explain the growth of



(1) Root elongating at a constant rate

(2) Early embryo development in an angiosperm

(3) Living organism growing in a natural environment

(4) Insect during rainy season

101. The phytohormone that promotes nutrient mobilization and helps in the delay of leaf senescence, can show all the given physiological effects, **except**

(1) Production of new leaves, chloroplasts in leaves and lateral shoot growth.

(2) Promotion of the multinucleate condition in a cell

(3) Growth of adventitious shoot

(4) Overcoming apical dominance

102. Find the **correct** sequence of the developmental processes in a plant cell.

(1) Meristematic cell → Differentiation → Plasmatic growth → Death

(2) Meristematic cell → Plasmatic growth → Differentiation → Senescence

(3) Plasmatic growth → Senescence → Expansion → Death

(4) Differentiation → Plasmatic growth → Meristematic cell → Death

103. How many of the following tissues are formed by the process of dedifferentiation and redifferentiation respectively?

Cork cambium, inter-fascicular vascular cambium, primary xylem, cork, secondary phloem, Intercalary meristem, secondary cortex

(1) 2, 3

(2) 2, 4

(3) 4, 3

(4) 3, 2

104. Combination of which of the given plant hormones can be used to increase yield of pineapple?

(1) Auxin, ABA

(2) Ethylene, Dormin

(3) Auxin, Ethylene

(4) ABA, Ethylene

105. Select the **correctly** matched pair.

(1) Auxin – Apical dominance

(2) Ethylene – Delay in senescence

(3) Cytokinin – Suppress stem elongation

(4) ABA – Breaking of seed dormancy

106. Which of the following hormone has precursor, methionine?

(1) ABA

(2) Ethylene

(3) Auxin

(4) Cytokinin

107. Who amongst the following, first isolated auxin from the tips of coleoptiles of oat seedling?

(1) F. Skoog

(2) E. Kurosawa

(3) F.W. Went

(4) Miller

108. Which phytohormone is known as stress hormone?

(1) Cytokinin

(2) Gibberellin

(3) Abscisic acid

(4) Ethylene

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109. The most intensively studied form of gibberellin is

(1) GA_4

(2) GA_3

(3) GA_2

(4) GA_1

110. Three independent researchers reported the purification and chemical characterisation of three different kinds of inhibitors inhibitor-B, abscission-II and dormin. Later all three were proved to be chemically identical and named

(1) Kinetin

(2) Zeatin

(3) Abscisic Acid

(4) Gibberellins

111. Read the given statements and choose the **correct** option.
Statement (A): The presence of $Fe-S$ in complex I, II and III of mitochondrial ETS is a common feature of these complexes.
Statement (B): The first member of the Krebs' cycle is the raw material for the synthesis of gibberellins and amino acids.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Statement (A) is correct but statement (B) is incorrect

(4) Statement (A) is incorrect but statement (B) is correct

112. Growth, at the cellular level, is principally a consequence of increase in

(1) Number of cells

(2) Tissue formation

(3) Secondary growth

(4) The amount of protoplasm

113. In _____, difference in shapes of leaves can be observed if present in different habitats like air and water. Select the correct option to fill the blank.

(1) Cotton

(2) Buttercup

(3) Coriander

(4) Larkspur

114. Ethephon is

(1) Most widely used compound as a source of ABA

(2) Used in artificial ripening of fruits

(3) Used to promote male flowers in cucumber

(4) Used to inhibit flowering in mango

115. Coconut milk can be used to

(1) Promote root growth and root hair formation

(2) Promote senescence

(3) Induce formation of male flower

(4) Counteract apical dominance

116. The number of FAD^+ reduced through Krebs cycle for one glucose molecule is

(1) 6

(2) 4

(3) 2

(4) 1

117. Read the following statements and state them as **True (T)** or **False (F)**.
a. When the electrons pass from one carrier to another via complex I to IV in ETC, they are coupled to complex V for ATP production.
b. Two copper centres are present in the complex III of ETS.
c. F_1 is an integral membrane protein, that forms a channel through which protons cross the inner membrane of mitochondria.
d. Inner membrane of mitochondria is permeable to protons, in the region of ATP synthase.
Choose the **correct** option from the following.

(1) a(T), b(F), c(F), d(T)

(2) a(F), b(T), c(T), d(F)

(3) a(F), b(T), c(F), d(T)

(4) a(T), b(T), c(T), d(F)

118. In the respiratory ETS,

(1) Ubiquinone receives electrons from complex II and III

(2) NADH dehydrogenase complex directly transfers its electrons to a mobile electron carrier on the outer surface of inner mitochondrial membrane

(3) Complex I and complex IV transfer hydrogen ions from the intermembrane space to the matrix of mitochondrion.

(4) A complex containing cytochromes and two copper centres passes electrons to O_2, resulting in the formation of H_2O

119. What will be the net gain of ATP, NADH + H^+ and $FADH_2$ during Krebs' cycle if one molecule of pyruvic acid is oxidised?

(1) 2 ATP, 6 NADH + H^+, 3 $FADH_2$

(2) 1 ATP, 6 NADH + H^+, 2 $FADH_2$

(3) 2 ATP, 6 NADH + H^+, 2 $FADH_2$

(4) 1 ATP, 3 NADH + H^+, 1 $FADH_2$

120. The **correct** sequence of electron carriers in electron transport chain of mitochondria is

(1) FMN → UQ → Cyt *b* → Cyt C_1 → Cyt *c* → Cyt *a*-a_3

(2) FMN → Cyt *b* → UQ → Cyt *a*-a_3 → Cyt *c* → Cyt C_1

(3) FMN → UQ → Cyt *b* → Cyt *c* → Cyt C_1 → Cyt *a*-a_3

(4) FMN → Cyt *a*-a_3 → Cyt *c* → Cyt C_1 → Cyt *b* → UQ

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121. If PEP is the respiratory substrate, then how many ATP molecules are produced in cytoplasm, in matrix and on inner membrane of mitochondria, respectively from its one molecule?

(1) 2, 2, 14

(2) 1, 15, 30

(3) 2, 2, 28

(4) 1, 1, 14

122. Which of the following is the favored substrate for respiration?

(1) Substrate formed by the degradation of invertase.

(2) Substrate that has the RQ value more than 1.

(3) Substrate that is the key product of deamination in mitochondria.

(4) Substrate that forms glucose-6-phosphate with the help of hexokinase and ATP.

123. Identify the following statements as **True(T)** or **False(F)** and choose the option accordingly.
A. Krebs' cycle was first elucidated by Hans Krebs, and the first member of the cycle is a 4 carbon compound.
B. TCA cycle must occur twice for each molecule of glucose respired.
C. Oxygen acts as terminal electron acceptor in ETS.

	A	B	C
(1)	T	F	T
(2)	T	T	F
(3)	T	T	T
(4)	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

124. Match the column I with column II and select the **correct** option.

	Column I		Column II
a	Aconitase	(i)	2-phosphoglycerate to phosphoenolpyruvate
b.	Enolase	(ii)	Citrate to isocitrate
c.	Decarboxylase	(iii)	DHAP to PGAL
d.	Isomerase	(iv)	Oxalosuccinic acid to α-ketoglutarate

(1) a(i), b(iii), c(iv), d(ii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iv), b(iii), c(i), d(ii)

125. Which of the following enzymes yields a five carbon compound?

(1) Hexokinase

(2) Aldolase

(3) Oxalosuccinic acid decarboxylase

(4) α-ketoglutaric dehydrogenase complex

126. Pyruvate dehydrogenase participates in

(1) Glycolysis

(2) Fermentation

(3) Krebs' cycle

(4) Link reaction

127. How many substrate level phosphorylation reactions will occur during Krebs' cycle for one sucrose molecule?

(1) Six

(2) Two

(3) Three

(4) Four

128. In prokaryotes, how many ATP molecules will be produced if two molecules of fructose-1, 6-bisphosphate undergoes its complete oxidation through aerobic respiration?

(1) 76

(2) 40

(3) 80

(4) 38

129. The complex IV of mitochondrial electron transport chain is also known as

(1) Cytochrome bc_1

(2) Cytochrome *c* oxidase complex

(3) Succinate dehydrogenase

(4) NADH dehydrogenase

130. If we take glycerol as respiratory substrate then the first molecule that enter the respiratory pathway is

(1) Acetyl CoA

(2) Pyruvic acid

(3) Phosphoglyceraldehyde

(4) 2-phosphoglycerate

131. The final hydrogen acceptor in electron transport system in mitochondria is

(1) Cytochrome *a*

(2) ADP

(3) Cytochrome *c*

(4) Oxygen

132. Which enzyme is responsible for the first irreversible reaction of glycolysis?

(1) Hexokinase

(2) Invertase

(3) Isomerase

(4) Enolase

133. The first member of Krebs' cycle is

(1) Succinyl CoA

(2) Oxaloacetic acid

(3) Citric acid

(4) Pyruvic acid

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134. How many carbon dioxide molecules are released in the Krebs' cycle from two molecules of Acetyl CoA?

(1) Two

(2) Six

(3) Four

(4) One

135. The mobile electron carrier towards the inter-membrane space of mitochondria is

(1) Ubiquinone

(2) Cytochrome bc_1

(3) Cytochrome *c*

(4) Cytochrome *a*

ZOOLOGY

136. Consider the following structures.
(A) Notochord
(B) Pharyngeal gill slits
(C) Dorsal hollow nerve cord
(D) Tail
Which of the above mentioned structures is/are present in adult Cephalochordates?

(1) Only (A)

(2) Only (A) & (B)

(3) Only (A), (B) & (C)

(4) (A), (B), (C) & (D)

137. An animal with a poison sting, aquatic life-style, two-chambered heart and exhibiting poikilothermy, is

(1) *Scoliodon*

(2) *Hippocampus*

(3) *Trygon*

(4) *Torpedo*

138. Select the **incorrect** option about sea walnuts.

(1) They exhibit bioluminescence

(2) Development is indirect

(3) Reproduction takes place only by sexual means

(4) Fertilisation is internal

139. Generally, chordates differ from non-chordates in presence of

(1) Nerve cord

(2) Heart

(3) Germ layers

(4) Pharyngeal gill slits

140. **Assertion :** Jelly fish exhibits metagenesis.
Reason : Jelly fish alternates between both polyp and medusa body forms.
In the light of above statements, choose the **correct** answer from the options given bellow.

(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

141. Select the correct match w.r.t classification of vertebrata.

(1) Vertebrata - Phylum

(2) Gnathostomata - Subphylum

(3) Pisces - Class

(4) Tetrapoda - Super class

142. Select the **correct** match.

(1) Tissue level of body organisation – *Sycon*

(2) Incomplete digestive tract – *Planaria*

(3) A viviparous aschelminth – *Enterobius vermicularis*

(4) Cells are arranged as loose cell aggregates – *Pennatula*

143. **Assertion (A) :** In larva of *Doliolum*, nerve cord is ventral to gut.
Reason (R) : Nerve cord in it is single and hollow.
In the light of above statements, choose the correct answer from the following options.

(1) (A) is true but (R) is false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

144. Read the following statements:
Statement A: All tetrapods are gnathostomes but all gnathostomes are not tetrapods.
Statement B: All gnathostomes are vertebrates but all vertebrates are not gnathostomes.
Choose the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

145. Choose the **incorrect** option w.r.t. birds.

(1) Presence of feathers is a characteristic feature

(2) Presence of scales on hind limbs show reptilian ancestry of birds

(3) Oil gland is present at the base of tail

(4) Air sacs connected to lungs are directly involved in gaseous exchange

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146. Identify a superclass among the following that includes organisms which bear fins for locomotion and choose the **correct** option.

(1) Cyclostomata

(2) Amphibia

(3) Gnathostomata

(4) Pisces

147. Choose the **odd** one w.r.t. characters of chordates.

(1) Nerve cord is dorsal, hollow and single

(2) Post anal tail is absent

(3) Pharynx perforated with paired gill slits

(4) Presence of Notochord

148. Among annelids, molluscs, arthropods, aschelminthes, platyhelminthes, echinoderms, hemichordates and chordates, how many have true coelom lined by mesoderm?

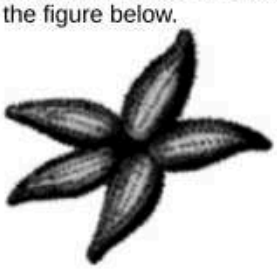
(1) Eight

(2) Six

(3) Seven

(4) Five

149. Choose the **incorrect** feature w.r.t. the organism shown in the figure below.



(1) Presence of water vascular system

(2) Monoecious

(3) Absence of excretory system

(4) Ventral mouth

150. Water vascular system in echinoderms does not help in

(1) Locomotion and respiration

(2) Transport of food

(3) Excretion and digestion

(4) Capture of food

151. Which of the following is/are living fossil(s) and belong(s) to the phylum Arthropoda?
(a) *Taenia*
(b) *Ctenoplana*
(c) *Limulus*
(d) *Dentalium*
Choose the correct option.

(1) (a) and (b)

(2) Only (c)

(3) (b), (c) and (d)

(4) (a), (b), (c) and (d)

152. Among the given characters in box, how many are **not** related to phylum Arthropoda?

Parapodia, Metamerism, Chitinous exoskeleton, Jointed appendages, Calcareous exoskeleton

(1) 4

(2) 3

(3) 2

(4) 1

153. **Assertion (A):** Feather-like gills are present in the mantle cavity of *Pila*.
Reason (R): Mantle cavity is the space between the hump and mantle.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

154. *Rana* and *Alligator* are similar to *Delphinus* and *Carcharodon* in having

(1) Bony endoskeleton

(2) Type of fertilisation

(3) Tympanum

(4) Gill slits in some stage of life

155. How many of the given organisms exhibit incomplete digestive system?
(a) *Euspongia*
(b) *Fasciola*
(c) *Hydra*
(d) *Ancylostoma*
(e) *Hirudinaria*
Choose the **correct** option.

(1) One

(2) Two

(3) Four

(4) Five

156. Select the **incorrect** match among the following.

(1) *Ancylostoma* — Dioecious, internal fertilisation

(2) *Pheretima* — Nephridia for respiration

(3) *Nereis* — Dioecious, possesses parapodia

(4) *Bombyx* — Metamerically segmented

157. How many of the following animals given below in the box are economically beneficial?

Apis, *Bombyx*, *Culex*, *Locusta*, *Pinctada*

Choose the correct option

(1) Two

(2) Three

(3) Four

(4) One

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158. What is true about *Balanoglossus*, *Sepia* and *Antedon*?

(1) All belongs to the same phylum

(2) All possess water vascular system

(3) All exhibit radial symmetry

(4) All possess true coelom

159. Select the option with **incorrect** statement w.r.t. Cyclostomata.
a. They have 6-18 pairs of gill slits for respiration.
b. Their larvae, after metamorphosis, return to fresh water.
c. Cranium and vertebral column are ossified.
d. Cyclostomes have terminal mouth with jaws.

(1) a and b only

(2) b, c and d only

(3) a, c and d only

(4) a, b, c and d

160. Read the following statements A and B. Choose the **correct** option.
Statement A : Members of the class chondrichthyes show internal fertilization and many of them are viviparous.
Statement B : Due to the absence of air bladder, cartilaginous fishes have to swim constantly to avoid sinking.

(1) Statement A is correct but B is incorrect.

(2) Statement A is incorrect but B is correct.

(3) Both statements A and B are correct.

(4) Both statements A and B are incorrect.

161. Select the **mismatch** w.r.t. animals and their respective phylum.

(1) Portuguese man of war – Cnidaria

(2) Brittle star – Echinodermata

(3) Tapeworm – Platyhelminthes

(4) Hookworm – Mollusca

162. Read the following statements and choose the option which **correctly** state them as true (T) or false (F).
A. *Locusta* possesses chitinous exoskeleton..
B. In molluscs, feather-like gills are present in mantle cavity.
C. Calcareous ossicles form exoskeleton in most of the echinoderms.
D. In *Branchiostoma*, notochord is present only in larval tail.

(1) A-(T), B-(F), C-(T), D-(F)

(2) A-(T), B-(T), C-(F), D-(F)

(3) A-(F), B-(T), C-(F), D-(T)

(4) A-(T), B-(T), C-(T), D-(F)

163. Closed circulatory system and metameric segmentation is the characteristic feature of

(1) *Nereis*

(2) *Limulus*

(3) *Aedes*

(4) *Anopheles*

164. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Metagenesis	(i)	*Pleurobrachia*
b.	Eight ciliated comb plates	(ii)	*Limulus*
c.	Book gills	(iii)	*Obelia*
d.	Lac insect	(iv)	*Laccifer*

(1) a(iii), b(i), c(ii), d(iv)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iv), b(i), c(ii), d(iii)

165. Mark the incorrect match.

(1) *Exocoetus* - marine - flying fish

(2) *Betta* - aquarium - fighting fish

(3) *Pterophyllum* - fresh water - stone fish

(4) *Torpedo* - marine - electric fish

166. Which of the following phyla exhibits organ system level of organisation, bilateral symmetry, presence of coelom, three germ layers and metameric segmentation?

(1) Annelida

(2) Platyhelminthes

(3) Aschelminthes

(4) Echinodermata

167. Presence of ctenoid/cycloid scales is a distinct characteristic feature of members of

(1) Super class Agnatha

(2) Super class Tetrapoda

(3) Class Osteichthyes

(4) Class Chondrichthyes

168. Which of the following animals is hermaphrodite?

(1) Spider

(2) *Euspongia*

(3) *Ascaris*

(4) Cockroach

169. A researcher found an organism 'X' near a sea shore. On careful analysis, he saw the presence of feather-like gills in 'X' within mantle cavity. Identify 'X' and choose the **incorrect** option w.r.t. the phylum to which 'X' belongs to.

(1) Organisms can be terrestrial or aquatic

(2) Organisms possess sensory tentacles at the anterior head region

(3) Specialised organ is present for respiration

(4) Organisms show metagenesis

170. Chondrichthyes differ from Osteichthyes as former possess

(1) Operculum

(2) Air bladder

(3) Placoid scales

(4) Two-chambered heart

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171. **Assertion (A):** In birds, both lungs and air sacs help in breathing.
Reason (R): Both lungs and air sacs are vascularized structures that provide the site for exchange of gases between blood and the atmosphere.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) correctly explains (A)

(2) Both (A) and (R) are true, (R) does not explain (A) correctly

(3) (A) is true; (R) is false

(4) Both (A) and (R) are false

172. Direct development is present in which animal?

(1) *Pheretima posthuma*

(2) *Periplaneta americana*

(3) *Rana tigrina*

(4) *Anopheles stephensi*

173. Tracheal tubes are respiratory structures of

(1) Insects

(2) Molluscs

(3) Coelenterates

(4) Annelids

174. How many of the features listed below is/are **correct** w.r.t. members of the phylum Porifera?
(i) Internal cavities are lined with food filtering flagellated cells
(ii) Body is made up of meshwork of cells
(iii) Mostly asymmetrical
(iv) Possess water vascular system

(1) Zero

(2) One

(3) Two

(4) Three

175. Identify the given animal and select the incorrect option regarding this animal.



(1) Absence of blood vessels

(2) Cranium and vertebral column are cartilaginous

(3) Presence of unpaired fins

(4) Shows indirect development

176. Water vascular system is not found in

(1) Sea urchin

(2) Sea lily

(3) Sea hare

(4) Sea cucumber

177. Which of the following is a common feature among birds and insects?

(1) Presence of ventral heart

(2) Presence of dorsal hollow nerve cord

(3) Absence of closed type of circulation

(4) Presence of crop and gizzard in alimentary canal

178. Match column I and column II. Choose the **correct** option.

	Column I		**Column II**
a.	*Balanoglossus*	(i)	Cyclostomes
b.	*Cucumaria*	(ii)	Air sacs
c.	*Petromyzon*	(iii)	Water vascular system
d.	*Neophron*	(iv)	Proboscis gland

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(i), d(ii)

179. Match column I and column II

Column I	**Column II**
a. Squid	(i) *Aurelia*
b. Star fish	(ii) *Lepisma*
c. Jelly fish	(iii) *Loligo*
d. Silver fish	(iv) *Asterias*

Select the **correct** option

(1) a(iv), b(i), c(iii), d(ii)

(2) a(ii) , b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

180. Choose the **correct** option to complete the analogy
Periplaneta : Malpighian tubules :: *Taenia* : _____

(1) Feather-like gills

(2) Coxal gland

(3) Nephridia

(4) Flame cells

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